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Sleep Health and Lifestyles
Exploratory Analysis

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1 Description of the data set

1.1 Problem

The prevalence of sleep disorders varies significantly, depending on the specific disorder. One of the most common sleep disorders is insomnia, which affects roughly one-third of adults. Though in general, about three-quarters of adults report having at least one sleep disorder symptom. The rate of diagnosis and treatment has nearly doubled over the past decade, indicating a growing concern. Poor sleep not only results in discomfort, but could also be linked to serious health consequences such as obesity, heart disease, diabetes and stroke. This highlights the importance of understanding the correlation between sleep health and lifestyle [2].

The dataset used in this project, obtained from Kaggle [1], provides information on lifestyle, health measures, and sleep-related variables for 374 individuals. Attributes include age, gender, occupation, BMI category, blood pressure, heart rate, physical activity level, daily steps, stress level, sleep duration, quality of sleep, and whether the individual has been diagnosed with a sleep disorder. The variety of attributes allows us to explore different sleep outcomes.

Previous analyses of this dataset show strong correlations between stress and sleep quality, and that physical activity and daily steps are associated with better sleep outcomes. One of the analysis reports suggested over 40% of participants were sleeping less than 7 hours per night. A huge divergence is seen between the sleep quality of individuals with high and low stress. Whereas, physical activity showcases a positive effect on sleep quality. The report highlights that "good sleep" is obtained by individuals with moderate stress levels and healthy physical activity.

1.2 Classification and Regression

In this project, we aim to use this dataset to analyse which factors, such as stress, physical activity, or daily habits, are most strongly associated with sleep quality and sleep disorders. Most particularly, we want to identify which attributes are the strongest predictors of sleep disorders.

For the regression task, it is important to choose an attribute that is both continuous and dependent. In this case, we would like to work with the "Sleep duration" variable as it directly reflects an individual's sleeping habits. Attributes like age can be directly used to capture the changes across the lifespan. Applying a regression model will help us identify the relationship between the variables, whether it's linear or complex. Though we already expect to see a negative relation between stress level and sleep duration, but a positive one if sleep duration is correlated with physical activity.

On the other hand, for the classification task, a categorical variable is more suitable, which is why we want to determine the sleep disorder. In our case, this attribute has three categories: None, Insomnia and Sleep Apnea. Quality of sleep and stress level are the strongest variables that can be used to predict the sleep disorder, as high stress is directly linked to insomnia. Using classifying methods, we want to see which features strongly separate people with and without sleep disorders.

Initial steps for preparing the data have already been taken. Numeric variables were standardised for PCA, and the categorical variables will be encoded before model training. Since the dataset contains no missing values, no imputation is required. The transformations ensure that the dataset will be suitable for both regression and classification models. Further transformation could be explored in the next project to enhance model performance.

2 Description of attributes in the data

2.1 Attributes

The dataset contains a mix of continuous, interval, and categorical attributes as seen in the table below.

Attribute	Type of Attribute	Unit
Person ID	Identifier	-
Age	Numeric (Continuous)	Years
Gender	Categorical (Nominal)	-
Occupation	Categorical (Nominal)	-
Sleep Duration	Numeric (Continuous)	Hours
Quality of Sleep	Interval (Discrete)	1 -10 rating
Physical Activity Level	Numeric (Continuous)	Minutes per day
Stress level	Interval (Discrete)	1 -10 rating
BMI Category	Categorical (Ordinal)	-
Blood Pressure	Numeric (Continuous)	mmHg
Heart Rate	Numeric (Continuous)	Beats per minute
Daily Steps	Numeric (Continuous)	Steps per day
Sleep Disorder	Categorical (Nominal)	-

Table 1: Overview of attributes

Overall, this mixture of attribute types makes the dataset well-suited for both regression and classification tasks after appropriate preprocessing.

2.2 Data Issues

The dataset used in this project is relatively clean. As already mentioned, there are no missing values. Though it still required some preprocessing:

- Person ID is removed, as it is just an identifier.
- Blood pressure is categorised according to the systolic and diastolic measurements.
- BMI was divided into four different categories: Normal, Normal weight, Overweight, and Obese. We fused Normal and Normal weight to avoid redundancy.
- Sleep disorder attribute originally contained three categories: None, Insomnia, and Sleep Apnea. The category "None" was mistakenly interpreted as a missing value (NaN) in Python. To correct this, we renamed the category to "No disorder".

2.3 Descriptive Statistics

To understand our data, we will provide an overview of its key characteristics. This initial exploration of both numeric and categorical attributes through a summary of their statistics, focusing on their tendencies and variation.

The following table presents the dataset, described through descriptive statistical analysis of the numeric attributes. The statistics show:

- The age of participants ranges from 27 to 59 years, with a mean of 42.2 and a median of 43, showing the dataset is primarily composed of middle-aged adults.

- The sleep-related and stress level measures show fascinating variation: sleep duration is distributed around seven hours (mean 7.13, std 0.8), while stress level (range 3–8, std 1.77) and Quality of sleep (range 4–9, std 1.2) display greater spread.
- Quality of sleep is slightly right-skewed, with a few higher values pulling the mean above the median.
- Physical activity and daily steps exhibit high variance relative to their range (std 20.8 and 1618).
- Daily steps also show mild left-skewness, with the mean 6817 below the median 7000, suggesting that some of the participants take fewer steps than the rest.
- Heart rate ranges from 65 to 86 bpm with a mean of 70.2 and a median of 69.0. This reflects some skewness, with some individuals having high values, making it the only attribute with potential outliers.

Figure 9 (see appendix) provides a visual illustration of the distributions and skewness, highlighting these characteristics more clearly. Furthermore, the potential presence of outliers will be further explored in Section 3.1, as they may have a significant influence on the performance of the predictive model

	Age	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	Heart Rate	Daily Steps
Count	374.0	374.0	374.0	374.0	374.0	374.0	374.0
Mean	42.18	7.13	7.31	59.17	5.39	70.17	6816.84
Std	8.67	0.8	1.2	20.83	1.77	4.14	1617.92
Min	27.0	5.80	4.0	30.0	3.0	65.0	3000.0
Q1	35.25	6.40	6.0	45.0	4.0	68.0	5600.0
Median	43.0	7.2	7.0	60.0	5.0	70.0	7000.0
Q3	50.0	7.80	8.0	75.0	7.0	72.0	8000.0
Max	59.0	8.50	9.0	90.0	8.0	86.0	10000.0

Table 2: Summary statistics of numeric attributes in the dataset

Variable / Category	Count	Percentage
Gender		
Male	189	50.5%
Female	185	49.5%
BMI Category		
Normal	216	57.8%
Overweight	148	39.6%
Obese	10	2.7%
Sleep Disorder		
No Disorder	219	58.6%
Insomnia	77	20.6%
Sleep Apnea	78	20.9%
Blood Pressure Category		
Normal	41	11%
Prehigh	262	70.1%
Stage 1 High	71	19%

Table 3: Frequency distribution of categorical attributes in the dataset.

To characterize the population of the dataset, Tables 3 and 4 present the frequency distribution of categorical variables. The gender distribution is the most well-balanced variable, with a nearly 50/50 split. On the other hand, BMI, Sleep Disorders, and Blood Pressure have less balanced distributions. With these categories, there is a subcategory that represents over half of the population (Normal weight, no disorder, Prehigh blood pressure), while the others make up a smaller portion. Furthermore, not all types of BMI, sleep disorders, and blood pressure are present. This imbalance may introduce bias, as the dominant subgroups are more strongly represented in the data, while the smaller subgroups provide limited information, making them harder for models to learn and predict accurately.

Occupation	Count	Percentage
Nurse	73	19.5%
Doctor	71	19.0%
Engineer	63	16.8%
Lawyer	47	12.6%
Teacher	40	10.7%
Accountant	37	9.9%
Salesperson	32	8.6%
Software Engineer	4	1.1%
Scientist	4	1.1%
Sales Representative	2	0.5%
Manager	1	0.3%

As for Occupation, it is dominated by Nurses, Doctors, and Engineers, who together make up 55 percent of the population. Other occupations, like Manager, Sales Representative, Software Engineer, and Scientist, are very underrepresented. These distributions are very important to keep in mind when going further into the processes of making a model and interpreting it.

Table 4: Frequency distribution of Occupation in the dataset.

3 Data visulization and PCA

This section continues the exploration and analysis of the dataset through visualizations and PCA: The visualizations provide an overview of distributions and relationships between variables, highlighting patterns and potential outliers. Building on this, PCA is applied to explore a possible reduction of dimensions, and shortly explore previously hidden patterns and structures.

3.1 Data visualization

As an extension to Section 2.3, Figures 1 and 9 (check appendix) illustrate the distribution of the attributes. Figure 9 includes countplots and histograms of all attributes, providing an overview of the dataset. As previously noted, there are imbalances in some categorical attributes, such as BMI and Blood Pressure. The skewness of the Heart Rate attribute is clearly visible, with most values clustering at the lower end and a few higher observations pulling the distribution to the right. However, none of the variables are strictly normally distributed. Figure 1, which shows boxplots of numeric attributes, further highlights this skewness and confirms the presence of outliers.

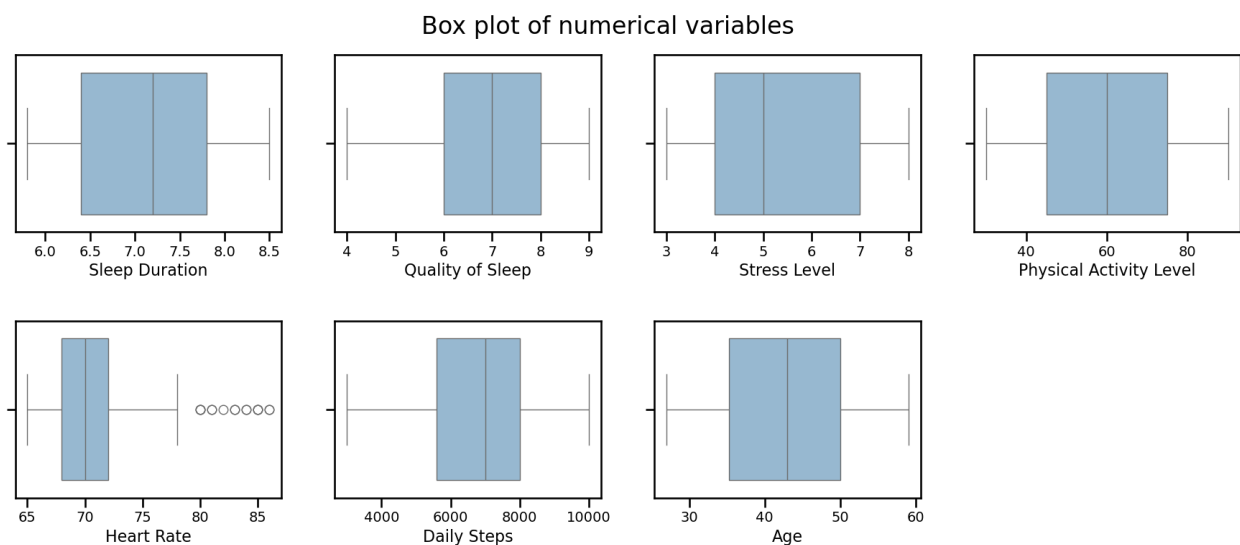


Figure 1

Secondly, we explore the connections between the attributes. The pairplots in Figure 2 provide a visual overview of the relations between the numerical variables, colored by sleep disorder categories. This allows patterns and groupings between the data to be observed. As a follow-up, a heatmap is also provided to get concrete numbers on the attribute correlations in Figure 3. One of the noteworthy connections visible is the linear connection between Sleep Duration and Quality of Sleep, Quality of Sleep and Stress Level, and, lastly, Daily Steps and Physical Activity Level. The correlations are also reflected in the numbers, where their correlation coefficients are 0.88, 0.9, and 0.77, respectively. Strong associations between sleep duration, quality of sleep, and stress suggest that these attributes capture related aspects of sleep health, making them relevant for predicting sleeping hours and classifying sleep disorders. Focusing on Sleep Duration, the lowest correlations are with Age, Activity Level, and Daily Steps, though their relation can resemble other nonlinear types. Age could be quadratically connected to Sleep Duration, and Daily Steps could be log- or sqrt-transformed. Activity Level does have good correlation for most of the data, except for two small clusters at the bottom right and top left, dragging down the correlation. Heart Rate also has medium correlation with Sleep Duration, but could also show inverse/decay relations. This could be an indication to try out some log or $1/x$ transformation later in the process to see a better fit.

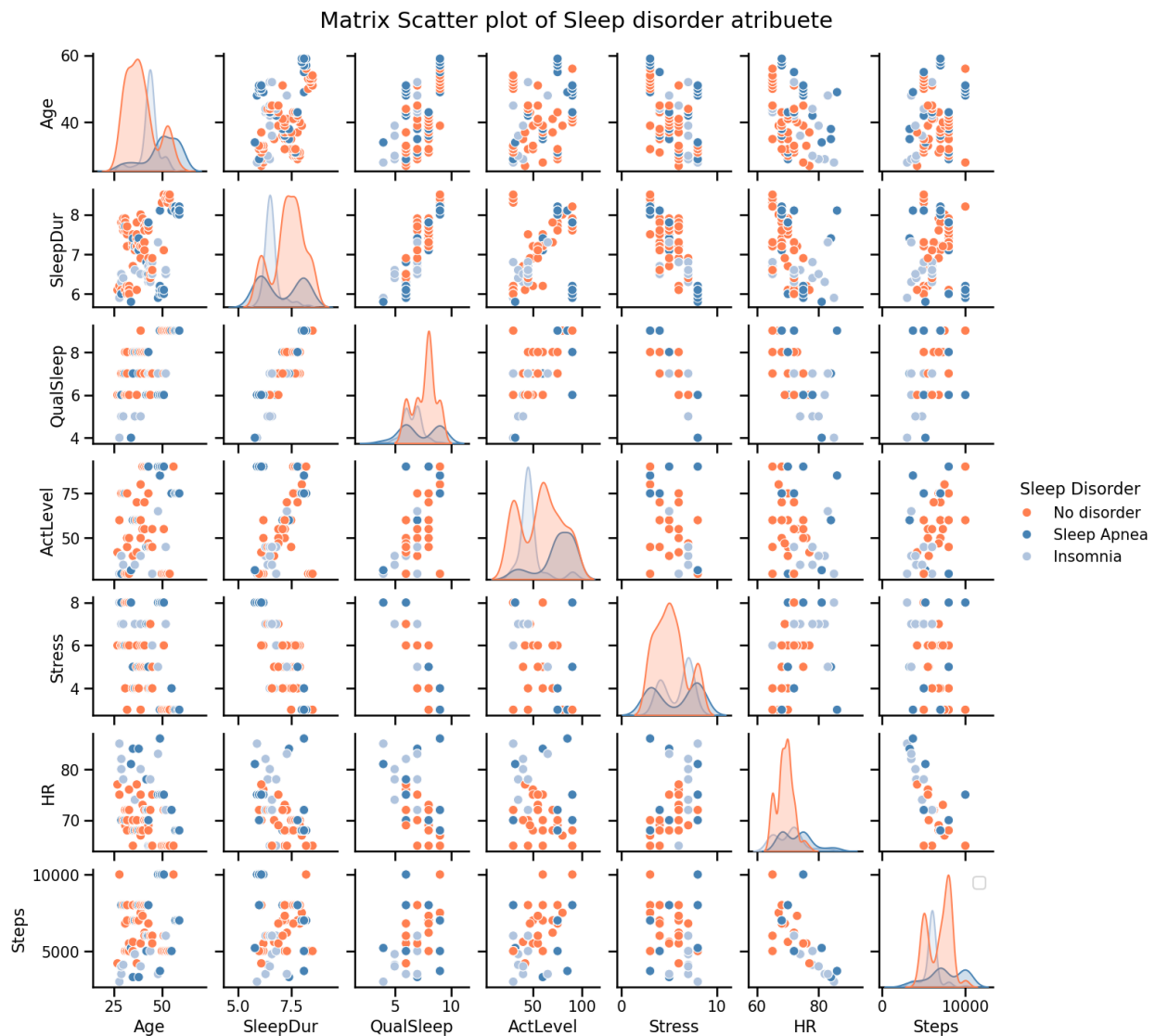
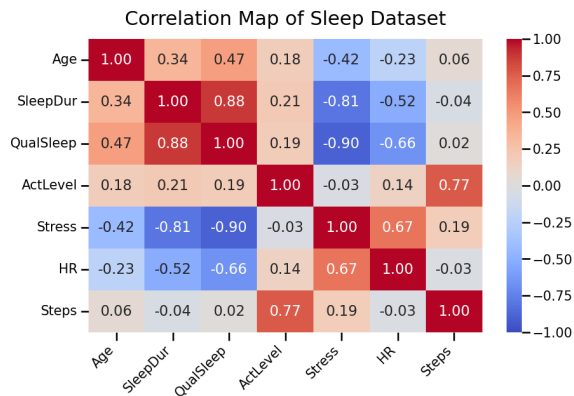


Figure 2



There are some interesting groupings of the disorders as well. A general trend appears, where the sleep disorders Sleep Apnea and Insomnia tend to group around or at the end of the No disorder cluster. This suggests that individuals without a disorder occupy a more central range of values, while the presence of a disorder is associated with more extreme patterns, which could be useful for distinguishing between groups in the classification task.

Figure 3

Since the connection between categorical variables could not be explored with a simple correlation matrix, we decided to explain it with distribution plots of sleep disorders by each category in Figure 4. Each subplot shows the proportion of individuals within each category that fall into the different sleep disorder groups. The following are the visible patterns:

- Females suffer more from sleep disorders (around 55%), whereas 70% of men do not suffer from any.
- Sleep disorders become more frequent as the BMI category goes up. An overwhelming majority in the normal category experience no sleep disorder, while everyone in the obese category suffers from a sleeping disorder.
- Higher blood pressure is also often associated with sleep disorders. Stage 1 high has an overwhelming majority with sleep apnea.
- Some professions (like teachers and nurses) show higher rates of disorders compared to others, while categories like engineer or accountant are dominated by no disorder.

The distribution of disorders across occupations can differ greatly between categories, but this may also be due to the limited number of data points in some of them. For example, the Manager category only contains 1 datapoint, and the Sales Representative only 2, which is why they only have one disorder each, and therefore can lead to less reliable conclusions.

Distribution of Sleep Disorders by Gender, BMI, Blood Pressure, and Occupation

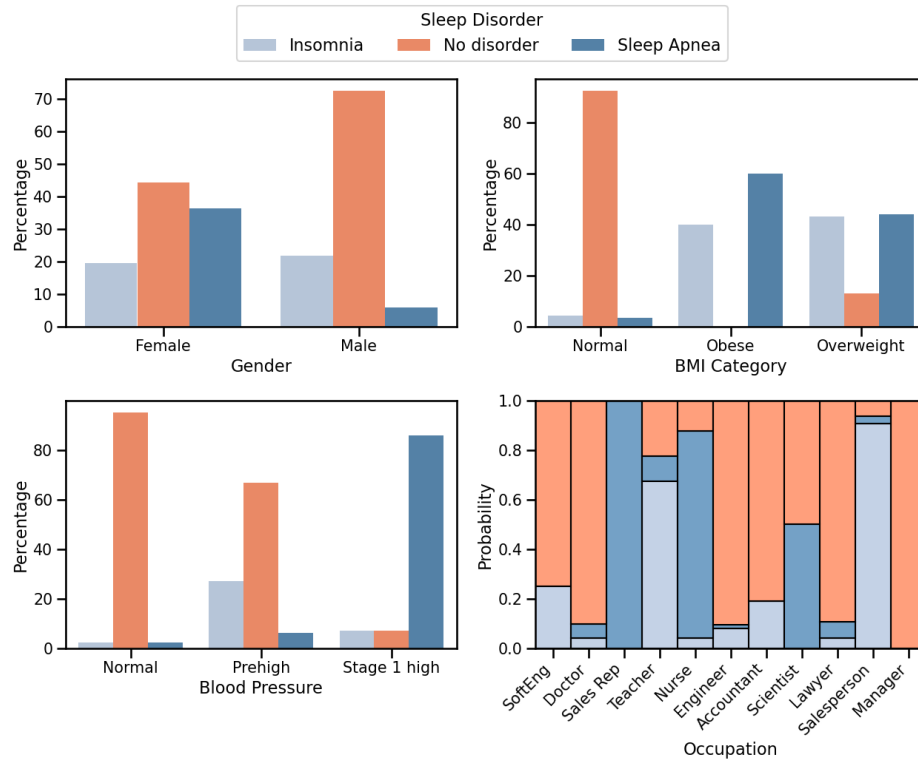


Figure 4

3.2 PCA

As part of our analysis, we applied Principal Component Analysis (PCA) in order to reduce the number of dimensions while still retaining as much of the explained variation as possible. Before applying PCA, we standardized the data to ensure that all attributes were on the same scale, since we see attributes with a single digit and others much larger. This would help avoid misleading results and give a clearer picture. In our dataset, we had 5 continuous attributes which correspond to 5 dimensions. PCA therefore computes five principal components (PC1-PC5), of which the first ones usually explain the largest proportion of the variation.

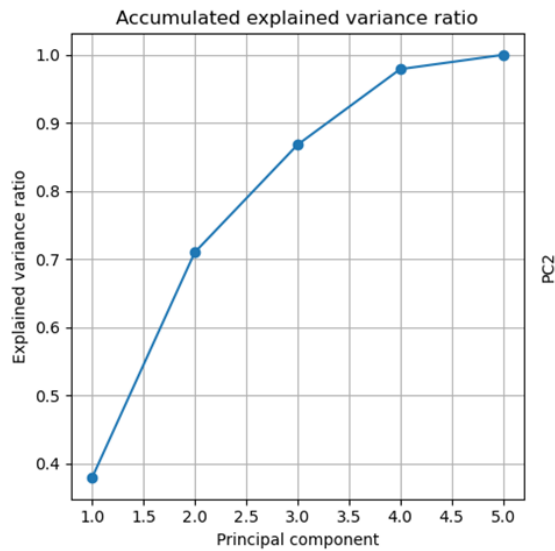


Figure 5

In Figure 5, we present the accumulated explained variance plot, which illustrates how much of the variance is explained by the principal components. PC1 and PC2 together explain most of the variation, around 72%. However, to reach approximately 98% of the explained variance, the first 4 components are required.

We wanted to see how each component affects each other; therefore, we plotted PCA pairwise, which resulted in 16 subplots (see Figure 10 in appendix). These plots reveal interesting combinations and patterns. In Figure 6, we can see that PC3 shows a negative relationship between some components, and PC4 exhibits some clustering among individuals with the same disorder. Although mathematically the first components explain the highest proportion of the variation, these plots really show how high-order components can still capture a meaningful structure in the data.

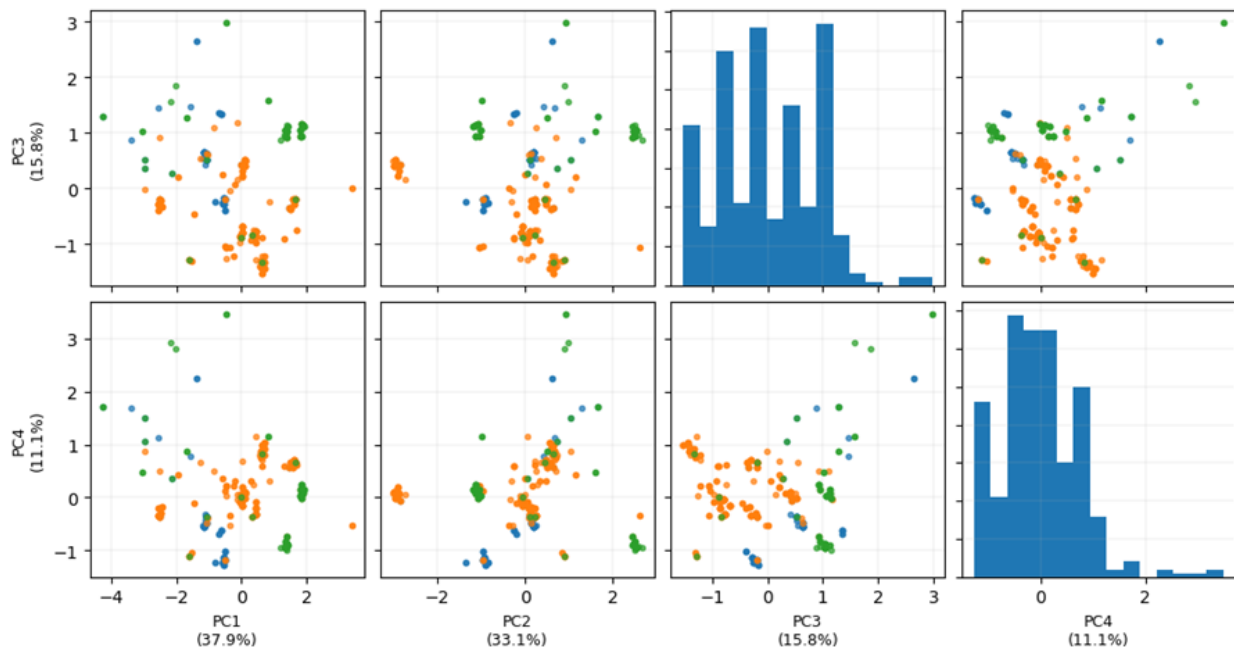


Figure 6: Pairwise scatterplots of PC3 and PC4 against the other principal components

Now, if we dig deeper into the first two components, as we can see in Figure ???. We can see which attributes contribute most to each component.

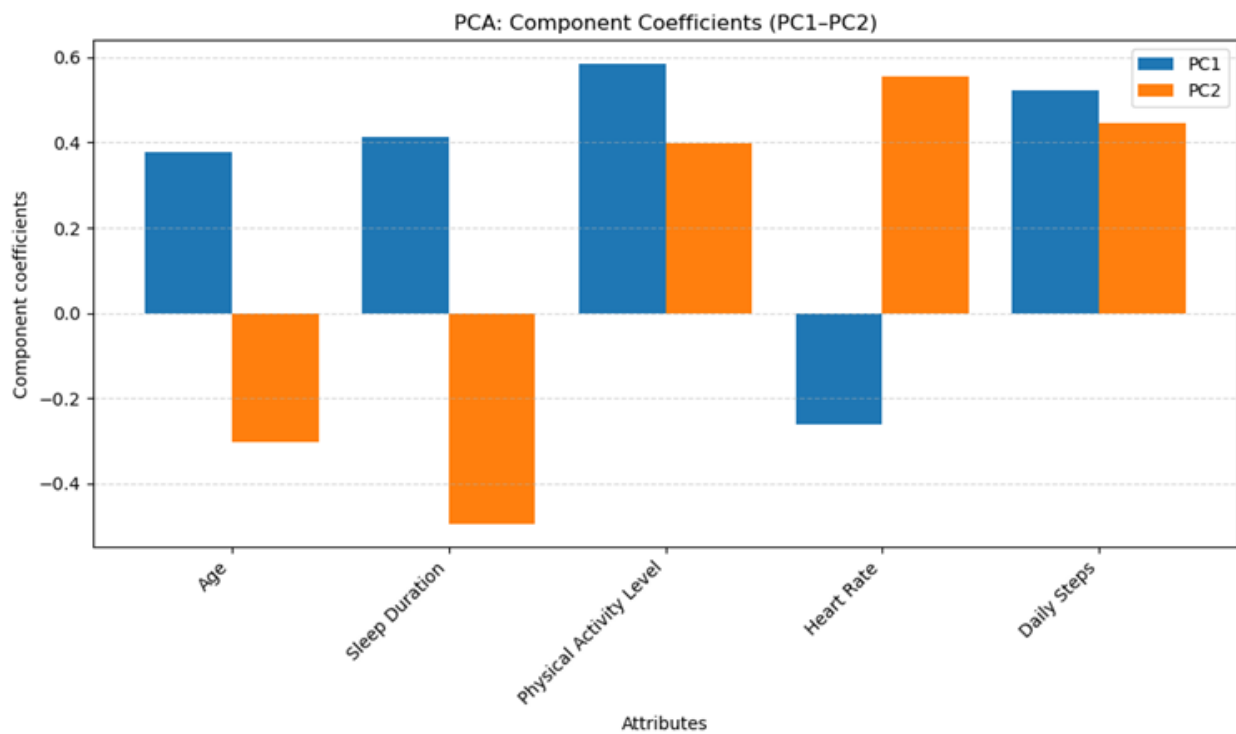


Figure 7

In PC1, the highest contributions come from daily steps and physical activity level, which both load positively, while heart rate contributes the least. For PC2, the strongest contributions are from heart rate and daily steps, both positive, while sleep duration also contributes strongly, but in the negative direction.

To look more closely at the connection between the first components, we projected the data in a 2D scatter plot and added a loading plot (see Figure 8). The scatter plot reveals that some groups tend to cluster according to their disorder, although there is still some overlap in the center. We also observe a few outliers that represent individuals who deviate from the main data structure.

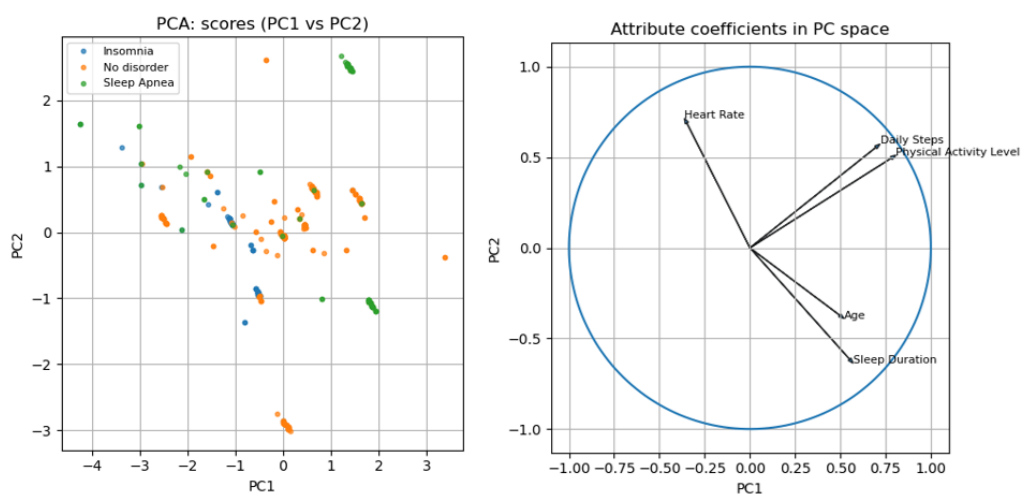


Figure 8

The loading plot further clarifies the relationships between the attributes. Daily steps and physical activity level are closely related, both loading strongly and in the same direction on PC1. Sleep duration and age are also in a similar direction, although age contributes less. This suggests that individuals with higher levels of physical activity and more daily steps generally have shorter sleep duration. Finally, heart rate contributes primarily to PC2, distinguishing it from the other attributes.

4 Discussion and Conclusion

Our analysis of the dataset shows both challenges and opportunities. The challenges we observed include outliers in one attribute, which could reduce predictive accuracy; therefore, we will remove them. There are also missing subcategories and dominating subcategories within some categorical variables, which can introduce bias. This is something that needs to be kept in mind when interpreting the final model and discussing its limitations. Finally, while some correlations with the target are low, the high correlations we do observe are clearly strong and provide solid starting points.

On the positive side, the dataset shows good overall variation, which is essential for building a model that is as representative of the real world as possible. There is a clear connection and pattern between the categorical attributes and Sleep Disorder, making classification appear feasible. This also includes patterns between Sleep Disorder and the numerical attributes. Moreover, Sleep Disorder is associated with a good number of correlated attributes, which motivates an exploration of PCA.

The purpose of the analysis was to reduce the number of dimensions while retaining as much explained variation as possible. PCA helped achieve this by reducing the components from 5 to 2 components, while still explaining 72% of the variation. If we wanted to retain almost 98% of the variation, the data could be reduced from 5 to 4 components. Although, this seems very inefficient to do so, since it offers little practical gain.

After comparing the option of keeping all five components with reducing to two, we decided to proceed with the latter. This choice allows us to simplify the model while still retaining most of the essential patterns and meaningful combinations, making further components unnecessary for our analysis.

Although the results regarding the first two components, were satisfactory, there are still points to discuss. Some overlap remained in the PCA plots, which made it difficult to separate the disorders. This could be because the data were projected onto only two dimensions, compressing variation that actually belongs in higher dimensions. It could also be due to shared characteristics among the disorder, which caused PCA to place them closer together.

While PCA was helpful, it also had limitations. PCA can only identify linear relationships, which means that more complex patterns such as quadratic or nonlinear connections may have been overlooked. In conclusion, PCA provided a useful reduction in dimensions and a clearer view of the data, but it also had limits.

5 Appendix

Feature Distribution Overview

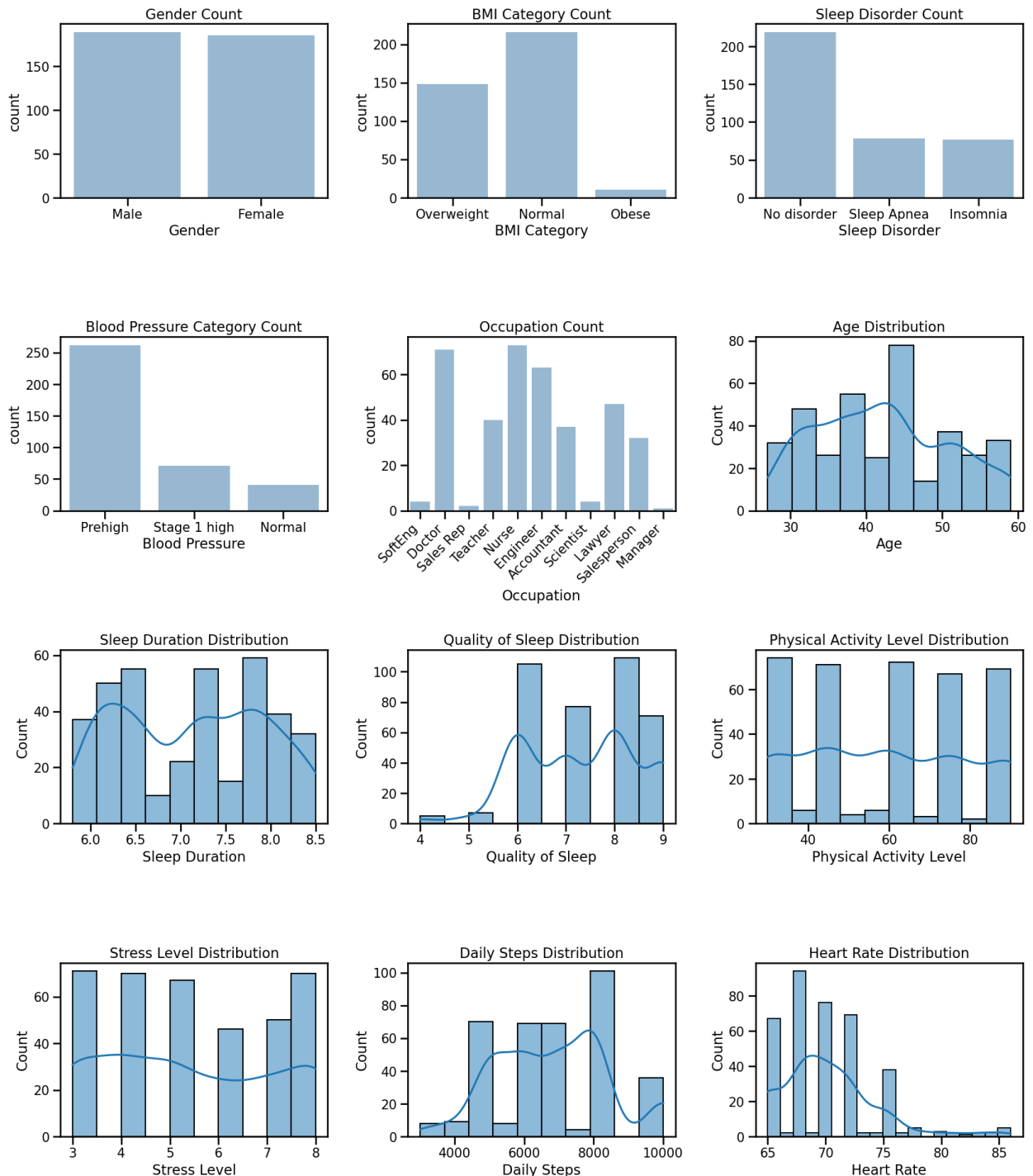


Figure 9

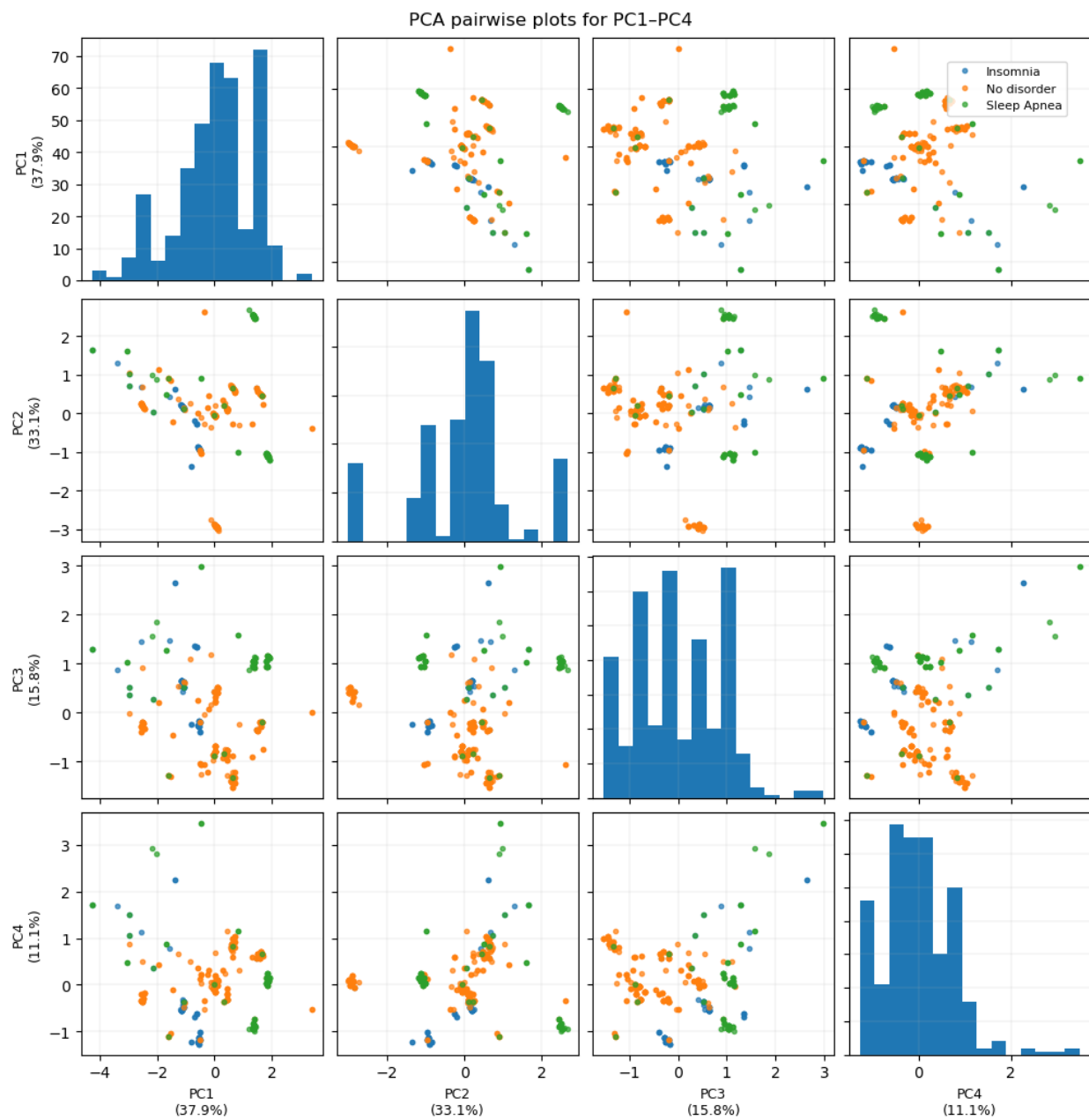


Figure 10

References

- [1] Kaggle. *Sleep Disorder Diagnosis Dataset*. 2023. URL: <https://www.kaggle.com/datasets/mdsultanulislamovi/sleep-disorder-diagnosis-dataset/data>.
- [2] Franziska Klimt et al. *Insomnia symptoms in adulthood: Prevalence and incidence over 25 years*. 2023. URL: <https://www.sciencedirect.com/science/article/pii/S1389945723002563>.
- [3] OpenAI. *ChatGPT*. ChatGPT has been used to formulate sentences in the report. 2025. URL: <https://chat.openai.com/>.